

Project Deliverables

The project aims to develop and implement a digital prescription, medication-tracking and inventory-management system for Bundoora Family Clinic. The clinic has relied heavily on manual processes, which have contributed to data-entry errors, inefficient record keeping, increased administrative workload and delays in retrieving information. The proposed system is intended to reduce these limitations by providing a secure, reliable and user-friendly digital solution.

From the beginning of the project, the system will be developed according to the identified functional and non-functional requirements, including FR1–FR45 and NFR1–NFR20. Based on the documented requirements and the clinic's operational background, the project will deliver a completed management system with an intuitive user interface, role-based access control and integrated modules for prescription, patient, appointment, clinic, testing, pharmacy and medication-inventory management.

The system will allow authorized users to create and manage digital prescriptions, track medication doses, receive reminders, update dispensing statuses and monitor medicine stock levels. It will also provide automated alerts for new prescriptions, missed doses, prescription renewals and low-stock medicines. In addition, users will be able to generate and export reports in PDF, Excel and CSV formats.

Security and access control will be important components of the proposed system. Role-based permissions will ensure that users can access only the functions relevant to their responsibilities. Administrators will manage user accounts, system settings and sensitive administrative functions, while doctors, patients and pharmacists will perform activities appropriate to their roles. This approach will reduce the likelihood of unauthorised changes, data misuse and system errors.

The project deliverables will include a data dictionary, entity relationship diagram, database design, system architecture, UI/UX wireframes and an interactive prototype. The project will also include a test plan, test cases, test data and test results to verify that the system meets its functional, usability, performance and security requirements. Testing will identify defects and confirm that corrective actions have been successfully implemented.

Overall, the proposed deliverables will support the complete development lifecycle of the system, including requirements analysis, design, development, testing, security implementation and deployment. The completed system is expected to improve data accuracy, reduce manual workload, increase the efficiency of prescription and inventory management, and provide Bundoora Family Clinic with a secure and accessible digital management solution